Estimation, Learning, and Stability Analysis of Supply Function Equilibrium Game for Generation Companies

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***Abstract*—In this paper, the competition among supplier agents in a uniform price electricity market is modeled as a supply function equilibrium game, where the players decide on a function of price versus quantity. The game problem is studied in two cases: in the first one, the players determine the intercept of the linear supply function; and in the second one, they choose its slope. It is assumed that the players do not have access to the cost function of the rivals and also cannot observe the current decision of them. Therefore, they estimate the future decisions of the rivals using their historical information. A nonlinear dynamic gradient learning method, namely myopic adjustment, is proposed for decision making of the players which works together with the estimation method. In each case, the Nash equilibrium point is analyzed and is shown that is a projection, onto a suitable subspace, of an equilibrium point of the dynamical system. Sufficient conditions for the stability of such equilibrium point of the proposed estimation and learning dynamics is obtained and an estimation of the corresponding region of attraction is computed using Lyapunov's second method.**

***Index Terms*—Dynamic game, generation companies, stability, supply function equilibrium (SFE).**

I. Introduction

THE concept of supply function equilibrium (SFE) was first introduced by Klemperer and Meyer [1] for modeling the profit maximizing behavior of the competing suppliers in a market with demand uncertainties. Starting from the work by Green and Newbery [2], the SFE model has been used in bidding strategy in electricity market, where the suppliers compete with each other to maximize their monetary payoffs by selling their electricity production to the market [3], [4].

Since, the SFE model is closely fitted to the real interaction among suppliers, especially in deregulated electricity market, analysts and industry advisers could use it as a modeling tool[5], [6]. It has to be considered that in a real electricity market, players have limited information about market data and rivals. When the bidding procedure is repeated in hour-ahead or day-ahead (DA) electricity market, the generating companies will be able to use the revealed historical information by independent system operator (ISO) and adjust their next bids in order to increase their profit [7]. Therefore, electricity market works as a dynamical system and it is important to construct an efficient bidding model to be consistent with dynamic nature of electricity market [8]–[10].

In [11], the linear SFE is utilized to model the strategic bidding strategies of suppliers. Only the intercept of the bid function is considered as the decision variable and a supplier estimates its rivals decisions through a linear gradient based dynamics. The optimal bidding strategy of the supplier is obtained by applying the linear quadratic Gaussian control, considering the risk factor in the payoff function of the suppliers. However, this study is not a game theoretical analysis and the problem is solved only from a single supplier's point of view. Recently, motivated by application of SFE in the wholesale market for the supply side, the supply function bidding model is also applied to the retail market for demand response in a smart energy grid [12]. The amount of demand willing to shed by a consumer is considered to be proportional to the price of electricity and the corresponding coefficient is considered as the strategy of a consumer. In addition, a distributed demand response algorithms is proposed to enable consumers to reach into the Nash equilibrium point.

In an electricity market, each supplier's payoff is affected by other suppliers' bid function. Therefore, it is more appropriate to model the interaction of the generation companies as a game problem [13]. In such an incomplete information repeated game, players need to have an estimate of the rivals' decision and adjust their decisions accordingly. Different adjustment techniques and different estimation methods can be used for this purpose like, naïve expectation and the best response adjustment[14], adaptive expectation and best response adjustment process [15], or adaptive adjustment [16], which can be used with naïve expectation [17] or adaptive expectation [18], simultaneously. The linear SFE game model have been studied where the suppliers use best response adjustment with naive expectation [19], [20]. In [19], it is assumed that players adjust the intercept of the linear supply function. Similar analysis for linear SFE game is given in [20], when the slope of supply function is considered as the strategy of suppliers. In the mentioned works, players decide fully rational based on their best response strategies and use the

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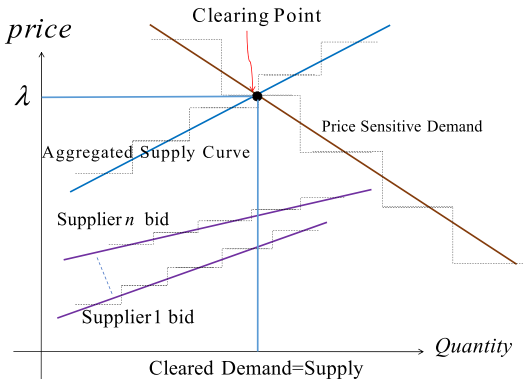

Fig. 1. Market clearing scheme.

naive estimation which leads to a linear dynamical bidding system. In addition, because of the complexity of the eigenvalues of the proposed linear dynamical system, they failed to give the conditions for the local stability of the Nash equilibrium point.

However, according to the fact that players decide in an uncertain environment, bounded rational decision making seems more reasonable [21], [22]. Players may use myopic adjustment based on profit gradient, that require less information and consequently a lower degree of rationality, where players adjust their decisions on the basis of the expected marginal profit [23], [24].

Therefore, players face a simultaneous dynamic estimation and decision making problem in the uncertain market environment. Now, the question is: under what conditions the estimation errors converge to zero and decisions converge to Nash Equilibrium point?

In this paper, a linear SFE model is presented and the competition among suppliers is modeled as a game in which:

1) the suppliers are the players of the game;
2) either intercept or slope of the supply function is the strategy of players;
3) the profit function of each player (i.e., supplier) is revenue minus cost.

It is assumed that demand is price sensitive and the players have limited information about other suppliers and also face an uncertain demand. To handle the lack of information and deal with market uncertainties, the players estimate the strategy of the rivals and adjust their strategy using a bounded rational decision rule, simultaneously. A continuous time adaptive expectation [18] together with continuous time myopic decision making method [24], [25] is used for this purpose. The Nash equilibrium points of the two games obtained using either intercept or slope of the supply function as the strategies of the players are derived, sufficient conditions for stability of the equilibrium points of the corresponding dynamical systems are achieved and the regions of attraction computed. The contributions of the paper could be summarized as follows.

1) Modeling incomplete information dynamic bidding strategies of suppliers for a linear SFE game using a nonlinear myopic adjustment and adaptive estimation.
2) Finding the closed form solution of Nash equilibrium point of the SFE game when either intercept or slope of the supply function is the strategy of the players. The conditions for existence and uniqueness are also given.
3) Providing the conditions for local stability of the nonlinear dynamic bidding system for both of the mentioned games.
4) Estimating the domain of attraction of the proposed nonlinear dynamical system using Lyapunov method.
5) Studying the effect of demand uncertainties on the convergence of the system and domain of attraction.

The rest of the paper is organized as follows: Section II provides the problem formulation and the system model; in Section III the game theoretic analysis, including the derivation of the Nash equilibrium point, and the analysis of the associated dynamical system are provided for the intercept strategies. The analysis is extended to the case of uncertain demand as well. The same concepts are studied when the slope of the supply function is the players' strategies in Section IV. In Section V, simulation results for an example are provided. Finally Section VI reports some conclusions.

II. System Model and Problem Formulations

Consider an oligopolistic electricity market composed of n suppliers trying to maximize their profit. It is assumed that the strategy of the players is their supply offering function which relates the price to the quantity. Each supply function indicates the level of quantity that a supplier is willing to sell at different prices. The market is considered to be a uniform price market like many DA electricity markets, e.g., California ISO (CAISO).

Parametrized linear supply function is a justified and applicable model of decision making of suppliers in electricity market [26]. Fig. 1 shows that the real piecewise supply and demand curves which are used in most of the electricity markets, can be approximated by some linear functions, so that the conditions of optimality can be derived and and equilibrium point of the game can be analyzed explicitly. It is assumed that the supply function of each company is in the following affine form:

$$Q_i = \eta_i(\lambda_i - \beta_i), \qquad i = 1, 2, \ldots, n \tag{1}$$

where λ_i and Q_i are the price and quantity variables of the ith player, respectively; η_i and β_i are positive coefficients. In this paper, it is assumed that the price is the same for all players, i.e., $\lambda_i = \lambda$ for all i.

The market demand is considered to be price sensitive as follows:

$$\text{Demand} = D_0 - \vartheta\lambda \tag{2}$$

where D_0 and ϑ are known positive constants. In a stable market, the amount of demand and supply must be equal. Therefore, by incorporating the supply functions of all players and meeting with market demand, a uniform price is cleared for all players as follows:

$$\text{Demand} = \sum_{i=1}^{n} Q_i$$

$$D_0 - \vartheta\lambda = \sum_{i=1}^{n} \eta_i(\lambda - \beta_i) = \lambda \sum_{i=1}^{n} \eta_i - \sum_{i=1}^{n} \eta_i\beta_i.$$

Hence,

$$\lambda = \frac{D_0 + \sum_i \eta_i\beta_i}{\vartheta + \sum_i \eta_i}. \tag{3}$$

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Above and in the rest of the paper, we omit the limits of the summation (from 1 to n) leaving only the index variable.

Fig. 1 shows the graphical interpretation of electricity market clearing mechanism. We see that the total supply curve is built up by aggregation of single supply curves of the generation companies. The intersection of the aggregated supply curve and the price sensitive demand curve gives the uniform market clearing price and the total amount of supply/demand. Once the price is cleared each supplier can realize the amount of its production from (1).

The profit function of the ith player is defined as follows:

$$\pi_i = Q_i\lambda - C_i(Q_i), \qquad i = 1, 2, \ldots, n \tag{4}$$

where $C_i(\cdot)$ is the quadratic cost function of the ith player described by

$$C_i(Q_i) = a_i Q_i^2 + b_i Q_i + c_i \tag{5}$$

with $a_i, b_i, c_i \geq 0$. Note that the above quadratic cost function satisfies the requirements of a cost function in microeconomic theory [27] and, having a linear marginal cost, it is also compatible for the class of linear supply functions [26].

As it is clear from above, each player's payoff is affected by other players' decisions through the market clearing price, which results in having a game problem rather than an optimization.

We model the behavior of suppliers within the following strategic (noncooperative) game framework. Let $\mathcal{N}$ be the set of n competing suppliers. Each supplier-i offers its supply curve characterized by the variables η_i or β_i. Since changing both intercept and slope rarely leads to a unique solution [28], [29], it is assumed that only one of them is changed and the other one is constant. In Sections III, we assume that η_i is some positive fixed constant and β_i is the strategy of the supplier-i; then, in Section IV, the parameters η_i are variable and β_i are constant. These two SFE games can be formalized as follows:

$$\mathcal{G}_1 := \{\mathcal{N}, \{\beta_i\}, \{\pi_i\}\} \quad \mathcal{G}_2 := \{\mathcal{N}, \{\eta_i\}, \{\pi_i\}\}. \tag{6}$$

In a real competitive electricity market, players do not have information on rivals' future decisions (i.e., β_i in $\mathcal{G}_1$, η_i in $\mathcal{G}_2$). Therefore, each player tries to maximize his/her payoff by taking into account the expected decisions of the rivals. Since the players don't have even the information of rivals' decision function types, in this paper it is assumed that each player uses the so called adaptive expectation method [18]—a model free estimation—to predict future decisions of the rivals as follows (we omit the time dependence of functions when this is unlikely to cause confusion):

$$\dot{\omega}_{ij} = \delta_{ij}(\omega_j - \omega_{ij}) \tag{7}$$

where $\omega_{ij}(t)$ is the estimated decision of the jth player by the ith player at time t, and $\delta_{ij} > 0$ is the constant learning rate of the i-th player to estimate the jth player.

Remark 1: Notice that we are assuming a continuous time setting as a limit approximation of a repeated game which is in fact in a discrete time setting. In such discrete time setting, information on other players' strategies cannot directly be used by one's decision since they are received after a delay equal to a time-step. In the continuous time limit, they are received at the same time of the decision but cannot be directly used anyway.

In $\mathcal{G}_1$, $\omega = \beta$; in $\mathcal{G}_2$, $\omega = \eta$; we will not change the name of the learning rate. It is assumed that the players adjust their strategy based on their local estimate of the profit gradient. This approach is known as a myopic adjustment [23], [24] in which, the player will increase (decrease) its strategy if the gradient of the payoff is positive (negative). This bounded rational decision making method works with weaker assumptions (since it works based on estimate of expected marginal profit) than the best response method [24]. In the myopic adjustment, the strategy of each supplier is updated in the direction of the gradient of profit function as

$$\dot{\omega}_i = \gamma_i \omega_i \frac{\partial \pi_i}{\partial \omega_i} \tag{8}$$

where $\gamma_i \geq 0$ is the constant factor of the learning rate and ω_i is the decision of the ith player at time t. (Here again we will not change the name of the learning rate when dealing with problems $\mathcal{G}_1$ and $\mathcal{G}_2$). Equation (8) means that the learning rate increases with the strategy of the player [24], [25]. This leads to a larger increase of the offered price for a supplier with higher strategy, which has greater capacity to increase (decrease) the price in the presence of a positive (negative) marginal profit. In fact, the right-hand side of (8) cannot be computed by the players because the profit $\pi_i = \pi_i(\omega_1, \ldots, \omega_n)$ and each player i knows exactly only ω_i. However, assuming the estimation (7) works well, we can approximate (8) through

$$\dot{\omega}_i = \gamma_i \omega_i \frac{\partial \pi_i}{\partial \omega_i}(\omega_{i*}) \tag{9}$$

where $\omega_{i*} = (\omega_{i1}, \ldots, \omega_{\text{in}})^T$ and $\omega_{ii} = \omega_i$.

Notice that the region $\omega_i \geq 0$ is invariant for (9), a necessary and sufficient condition to keep the bidding price of the suppliers positive for any quantity. Further, notice that the equilibria of (7), (9) are the same of (7), (8), i.e., the approximation is exact at equilibrium points.

In the next sections, we compute the Nash equilibrium points of the proposed SFE game model. In addition, the convergence of the dynamic estimation and decision making model (7), (9) to the true Nash strategies of the players is studied and a domain of attraction of the equilibrium point is computed. Moreover, thanks to the concept of practical stability, the effect of demand uncertainties is analyzed as well.

III. Analysis of the Game $\mathcal{G}_1$

A. Nash Equilibrium Point of $\mathcal{G}_1$

In the Nash equilibrium of the noncooperative game, the strategy of each player is the best response to the other players' policies. Therefore, the Nash equilibrium point will be at the intersection point of the all players' best responses. As a result, no player can achieve higher profit by independently changing her strategy.

Proposition 1: The game $\mathcal{G}_1$ generically admits a unique Nash equilibrium point, say $N^\beta \in \mathbb{R}^n$.

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Proof: According to the first derivative of the payoff functions (given in the Appendix, Section A) and by taking the second derivative of the ith player's payoff function we have

$$\frac{\partial^2 \pi_i}{\partial \beta_i^2} = \eta_i \left[-2a_i \eta_i \left(q_i - 1\right)^2 + 2\left(q_i - 1\right) q_i \right] \tag{10}$$

where

$$q_i = \frac{\eta_i}{\vartheta + \sum_j \eta_j}.$$

It is easily seen that $0 < q_i < 1$. Therefore, the second-order condition holds. The Nash equilibrium strategy of $\mathcal{G}_1$ is obtained as

$$N^\beta = Y^{-1} w. \tag{11}$$

The expressions of the matrix Y and the vector w are given in the Appendix, Section B; Y is an invertible matrix in almost all cases; therefore, the Nash equilibrium is unique in almost all the cases. ■

B. Dynamic Estimation and Learning of $\mathcal{G}_1$

By putting (60) into (9) and including (7), the dynamics of estimation and decision making based on adaptive expectation and myopic adjustment is as follows:

$$\dot{\beta}_i = \sum_{j \neq i} \gamma_i h_{ij} \beta_i \beta_{ij} + \gamma_i f_i {\beta_i}^2 + \gamma_i m_i \beta_i \tag{12a}$$

$$\dot{\beta}_{ij} = \delta_{ij}(\beta_j - \beta_{ij}), \qquad i, j = 1, 2, \ldots, n; \quad i \neq j \tag{12b}$$

where the constants h_{ij}, f_i, and m_i are defined in the Appendix, Section C. Further, let

$$\beta = (\beta_{1*}^T \ldots \beta_{n*}^T)^T \in \mathbb{R}^{n^2} \tag{13}$$

where we used on β_{i*} the notation introduced for ω_{i*} before.

Proposition 2: Corresponding to N^β the dynamics system (12) admits an equilibrium point $\beta^* = \underline{1}_n \otimes N^\beta \in \mathbb{R}^{n^2}$.

Proof: Since (12a) comes from (9), it is easily seen that an equilibrium point of (12) is

$$\gamma N_i^\beta \left. \frac{\partial \pi_i}{\partial \beta_i} \right|_{N^\beta} = 0, \qquad i = 1, \ldots, n$$

$$\beta_{ij}^* = \beta_j^* = N_j^\beta, \qquad i, j = 1, \ldots, n; \quad i \neq j$$

and the statement follows straightforwardly. ■

In the following, we will use again $\beta_i^* = N_i^\beta$.

C. Stability Analysis of $\mathcal{G}_1$

The estimation and decision dynamics in (12) can be written as a dynamical system of order n^2

$$\dot{\beta} = C\beta + \mathrm{diag}(\beta)\, B\, \beta. \tag{14}$$

The matrices C and B of dimensions $n^2 \times n^2$ can be constructed by composing $n \times n$ blocks C_{ij}, B_{ij} of dimensions $n \times n$, conformally to vectors β_{i*}, β_{j*} in (13). In turn, we indicate the elements of blocks C_{ij}, B_{ij} conformally with the elements of the above vectors, i.e., the pair (il, jm) will point at the intersection of il-row and jm-column

$$C_{ii} = \begin{array}{c} \\ (i,1) \\ (i,2) \\ \vdots \\ (i,i) \\ \vdots \\ (i,n) \end{array} \begin{array}{c} \begin{array}{cccccc} (i,1) & (i,2) & \ldots & (i,i) & \ldots & (i,n) \end{array} \\ \begin{pmatrix} -\delta_{i1} & 0 & \ldots & 0 & \ldots & 0 \\ 0 & -\delta_{i2} & \ldots & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \ldots & \ldots & \gamma_i m_i & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \ldots & 0 & \ldots & -\delta_{\mathrm{in}} \end{pmatrix} \end{array} \tag{15a}$$

C_{ij} with $i \neq j$ has all zero elements except for $C_{ij,jj} = \delta_{ij}$ (15b)

$$B_{ii} = \begin{array}{c} \\ (i,1) \\ (i,2) \\ \vdots \\ (i,i) \\ \vdots \\ (i,n) \end{array} \begin{array}{c} \begin{array}{cccccc} (i,1) & (i,2) & \ldots & (i,i) & \ldots & (i,n) \end{array} \\ \begin{pmatrix} 0 & 0 & \ldots & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \gamma_i h_{i1} & \gamma_i h_{i2} & \ldots & \gamma_i f_i & \ldots & \gamma_i h_{\mathrm{in}} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \ldots & 0 & \ldots & 0 \end{pmatrix} \end{array} \tag{15c}$$

$B_{ij} = 0$ for $i \neq j$. Now, letting

$$x(t) = \beta(t) - \beta^* \tag{16a}$$

$$\Sigma(x) = \mathrm{diag}(x) \tag{16b}$$

and remembering that $0 = C\beta^* + \Sigma(\beta^*)\, B\, \beta^*$, we have

$$\dot{x} = [\Sigma(\beta^*)\, B + C]\, x + \Sigma(x)\, B\, x + \Sigma(x)\, B\, \beta^*.$$

Finally, using the fact $\Sigma(x)\, B\, \beta^* = \Sigma(B\beta^*)\, x$, we get the following representation:

$$\dot{x} = Ax + \Sigma(x) B x \tag{17}$$

where

$$A = \Sigma(\beta^*)\, B + C + \Sigma(B\beta^*) \tag{18}$$

is the system matrix of the linearization of (17) around zero. The following result shows there exist values of the parameters ensuring stability of (17).

Theorem 1: There exists $\delta_{ij}, \gamma_i > 0$, $i, j = 1, 2, \ldots, n$, $i \neq j$ such that the matrix A is asymptotically stable (and hence system (17) is locally asymptotically stable), if the following inequality holds:

$$m_i + 2\beta_i^* f_i + \beta_i^* \sum_{j \neq i} (h_{ij} + |h_{ij}|) < 0, \quad i = 1, 2, \ldots, n. \tag{19}$$

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Proof: Consider the following permutation of the state variables:

$$\begin{aligned}\tilde{\beta}_{i*} &= \left[\beta_{ii}, \beta_{i1}, \ldots, \beta_{i(i-1)}, \beta_{i(i+1)}, \ldots, \beta_{\text{in}}\right]^T \\ &= \begin{bmatrix} \mathbf{0}_{1\times(i-1)} & 1 & \mathbf{0}_{1\times(n-i)} \\ \mathbf{I}_{(i-1)} & \mathbf{0}_{(i-1)\times 1} & \mathbf{0}_{(i-1)\times(n-i)} \\ \mathbf{0}_{(n-i)\times(i-1)} & \mathbf{0}_{(n-i)\times 1} & \mathbf{I}_{(n-i)} \end{bmatrix} \beta_{i*} \\ &= T_i \beta_{i*}.\end{aligned}$$

Then, stacking the state variables, we have

$$\begin{aligned}\tilde{\beta} &= \left(\tilde{\beta}_{1*}^T, \tilde{\beta}_{2*}^T, \ldots, \tilde{\beta}_{n*}^T\right)^T \\ &= \text{blockdiag}\left(T_1, T_2, \ldots, T_n\right) \left(\beta_{1*}^T, \beta_{2*}^T, \ldots, \beta_{n*}^T\right)^T \\ &= T\beta \end{aligned} \tag{20}$$

where clearly T is invertible. Now, if we transform the matrix A in (18) by T, we have

$$\begin{aligned}\tilde{A} &= T^{-1}\left[\Sigma(\beta^*)\, B + C + \Sigma(B\beta^*)\right] T \\ &= \Sigma(\beta^*)\, T^{-1} BT + T^{-1} CT + \Sigma(B\beta^*) \\ &= \Sigma(\beta^*)\, \tilde{B} + \tilde{C} + \Sigma(B\beta^*).\end{aligned} \tag{21}$$

Therefore, we just need to transform the matrices B and C to calculate $\tilde{A}$ according to (21). Since the eigenvalues of the matrices A and $\tilde{A}$ are the same, in what follows Gershgorin's circle theorem is used to find sufficient conditions guaranteeing all the eigenvalues of matrix $\tilde{A}$ to be negative. Consider $\text{disc}(o_k, r_k)$ as a disc in the complex plane centered at o_k with radius equal to r_k for $k = 1, 2, \ldots, n^2$. Let the centers o_k be given by the diagonal elements of the matrix $\tilde{A}$ and the radii r_k be given by the sum of the absolute values of the elements in the corresponding columns (except the diagonal element). In this way, two types Gershgorin's discs can be determined according to the entries of matrix $\tilde{A}$

$$\textbf{type 1}: \quad \text{disc}(\gamma_i \psi_i, \sum_{j\neq i} \delta_{ij}), \quad i = 1, 2, \ldots, n \tag{22}$$

$$\textbf{type 2}: \quad \text{disc}(-\delta_{ij}, |\gamma_i \beta_i^* h_{ij}|), \quad i, j = 1, 2, \ldots, n; \quad i \neq j \tag{23}$$

where

$$\psi_i = m_i + 2\beta_i^* f_i + \sum_{j\neq i} \beta_i^* h_{ij}, \quad i = 1, 2, \ldots, n. \tag{24}$$

To guarantee that A is asymptotically stable, we can require Gershgorin's discs to be entirely in the left-half plane. To this aim, all their centers should be negative, which results in

$$\psi_i < 0, \qquad i = 1, 2, \ldots, n \tag{25}$$

and

$$\textbf{type 1}: \quad \sum_{j\neq i} \delta_{\text{ij}} < -\gamma_i \psi_i, \quad i = 1, 2, \ldots, n \tag{26a}$$

$$\textbf{type 2}: \quad \delta_{ij} > |\gamma_i \beta_i^* h_{ij}|, \quad i, j = 1, 2, \ldots, n; \quad i \neq j. \tag{26b}$$

Summing up inequalities (26b) for $j = 1, 2, \ldots, n,\ j \neq i$ and comparing with (26a) yields

$$\gamma_i \beta_i^* \sum_{j\neq i} |h_{ij}| < \sum_{j\neq i} \delta_{ij} < -\gamma_i \psi_i, \quad i = 1, 2, \ldots, n. \tag{27}$$

It is not difficult to realize that a necessary and sufficient condition for the existence of the learning rates γ_i and estimation rates δ_{ij} satisfying (27)—and, in turn, (26a) and (26b)—is

$$\psi_i < -\beta_i^* \sum_{j\neq i} |h_{ij}|, \quad i = 1, 2, \ldots, n. \tag{28}$$

Putting (28) and (25) together yields

$$m_i + 2\beta_i^* f_i + \beta_i^* \sum_{j\neq i} (h_{ij} + |h_{ij}|) < 0, \quad i = 1, 2, \ldots, n \tag{29}$$

which completes the proof. ■

In the sequel we will use the Euclidean 2-norm for vectors, unless explicitly noted, and the corresponding induced norm for matrices so that:

$$||A|| = (\lambda_{\max}(A^T A))^{1/2}.$$

Proposition 3: Assume the matrix A is stable, and consequently let G and H be positive definite matrices satisfying

$$A^T H + HA = -G.$$

Then the zero equilibrium point of the dynamical system (17) is asymptotically stable within the ellipsoid

$$\Lambda_r = \{x |\, x^T Hx \leq r\}, \tag{30}$$

provided $r > 0$ is chosen so that

$$\Lambda_r \subseteq \Gamma = \left\{ x \,\middle|\, \|x\| < \frac{\lambda_{\min}(G)}{2\|H\| \max_i \left(\gamma_i \sqrt{f_i^2 + \sum_{j\neq i} h_{ij}^2}\right)} \right\}. \tag{31}$$

Proof: See the Appendix, Section D. ■

D. Studying the Effect of Uncertain Demand Function

As mentioned before, the bounded rational decision making is more reasonable in an uncertain environment. On the other hand, it is a well-known practical fact that the demand is price sensitive, in the sense that it decreases when the price increases, but the behavior of demand is uncertain from the market participants' points of view, since they cannot have an exact forecast of demand when they decide on their strategies. The uncertain market demand can be modeled as follows:

$$\text{Demand} = D_0 - \vartheta\lambda + d \tag{32}$$

where d is a an uncertain quantity, possibly time-varying, and we assume $|d(t)| \leq \bar{d} \ll D_0$ for all t. One can see that this additive disturbance affects the decision making of players through (12). In this case, since D_0 becomes $D_0 + d(t)$, m_i in (12) becomes $m_i + M_i d(t)$ where

$$M_i = q_i^2 + q_i(q_i - 1)(1 - 2a_i \eta_i) \tag{33}$$

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and as a results the matrix C in (14) is perturbed as

$$\bar{C} = C + S\,d(t),$$

where S is a diagonal matrix with the same structure as C in (15) replacing δ_{ij} with 0 and m_i with M_i. We utilize the equilibrium point computed without uncertainty to shift the state variable as in (16a) yielding

$$\dot{x} = A\,x + \Sigma(x)\,Bx + S\,(\beta^* + x)\,d. \tag{34}$$

Of course, due to the uncertain term $S\,\beta^* d$, the system will not converge to zero. Therefore, we study the stability of (34) in the sense that, if the initial condition x_0 is sufficiently close to zero, then $x(t)$ is bounded to a closed set around the equilibrium point when time goes to infinity.

Proposition 4: Consider the dynamical system (34) and choose an $\epsilon > 0$ such that $U_\epsilon = \{x|\,\|x\| < \epsilon\} \subset \Lambda_r$. Then it is possible to computes values $d_0(\epsilon) > 0$ and $\upsilon(\epsilon, \bar{d}) \in (0, \epsilon)$ such that if $\bar{d} < d_0(\epsilon)$ then the trajectories of the dynamical system (34) starting at U_ϵ will converge to U_v. (This notion is also called practical stability [30].)

Proof: See the Appendix, Section E. ■

IV. Analysis of the Game $\mathcal{G}_2$

In this Section, we study the simultaneous estimation and learning of the suppliers in $\mathcal{G}_2$ (i.e., the parameters η_i for $i = 1, 2, \ldots, n$ are the strategy of the suppliers and the parameters β_i are some constants). It has been proved that an optimal constant value of intercept for linear supply function is the corresponding marginal cost coefficient (i.e., b_i) [31]; hence in this study we assume $\beta_i = b_i$.

A. Nash Equilibrium Point of $\mathcal{G}_2$

Proposition 5: $N^\eta = (N_1^\eta, N_2^\eta, \ldots, N_n^\eta)^T$ is a Nash Equilibrium point of $\mathcal{G}_2$ if and only if

$$N_i^\eta = \frac{\vartheta + \sum_{l\neq i} N_l^\eta}{1 + 2a_i(\vartheta + \sum_{l\neq i} N_l^\eta)}, \qquad i = 1, 2, \ldots, n. \tag{35}$$

Proof: The profit function of the suppliers can be written as follows:

$$\begin{aligned}\pi_i &= \lambda\eta_i(\lambda - b_i) - a_i\eta_i^2(\lambda - b_i)^2 - b_i\eta_i(\lambda - b_i) - c_i \\ &= (\lambda - b_i)^2(\eta_i - a_i\eta_i^2) - c_i.\end{aligned}$$

Taking the first derivative of the profit function with respect to η_i results in

$$\begin{aligned}\frac{\partial \pi_i}{\partial \eta_i} =\ & (\lambda - b_i)^2(1 - 2a_i\eta_i) \\ & + 2(\lambda - b_i)\frac{b_l Z - D_0 - \sum_l \eta_l b_l}{Z^2}(\eta_i - a_i\eta_i^2),\end{aligned}$$

where

$$\frac{b_l Z - D_0 - \sum_l \eta_l b_l}{Z^2} = \frac{\partial \lambda}{\partial \eta_i} \quad Z = \vartheta + \sum_l \eta_l.$$

After more simplifications we obtain

$$\begin{aligned}\frac{\partial \pi_i}{\partial \eta_i} &= (\lambda - b_i)^2(1 - 2a_i\eta_i) - \frac{2}{Z}(\lambda - b_i)^2(\eta_i - a_i\eta_i^2) \\ &= -\frac{1}{Z}(\lambda - b_i)^2\left[-2a_i\eta_i^2 + 2(1 + a_i Z)\eta_i - Z\right].\end{aligned}$$

Now, considering $Z_i = Z - \eta_i = \vartheta + \sum_{l\neq i}\eta_l$, we have

$$\frac{\partial \pi_i}{\partial \eta_i} = -\frac{1}{Z}(\lambda - b_i)^2\left((1 + 2a_i Z_i)\eta_i - Z_i\right). \tag{36}$$

Therefore, the first order condition results in

$$\lambda - b_i = 0 \text{ or } (1 + 2a_i Z_i)\eta_i - Z_i = 0, \text{ for } i = 1, 2, \ldots, n. \tag{37}$$

Now, consider $N^\eta = [N_1^\eta, \ldots, N_n^\eta]^T$ as a solution for a set of n equations in (37). If $\lambda - b_k = 0$ for some player-k, by differentiating (36) we have

$$\left.\frac{\partial^2 \pi_k}{\partial \eta_k^2}\right|_{N^\eta} = 0. \tag{38}$$

Consequently, η^* cannot be the Nash equilibrium point of the game (it is a saddle point).

Therefore, a Nash equilibrium point must necessarily satisfy the following set equations:

$$\eta_i = \frac{Z_i}{1 + 2a_i Z_i}, \qquad i = 1, 2, \ldots, n. \tag{39}$$

By differentiation, the second-order condition results in

$$\left.\frac{\partial^2 \pi_i}{\partial \eta_i^2}\right|_{N^\eta} = -\frac{1}{Z}(\lambda - b_i)^2(1 + 2a_i Z_i) < 0,\ i = 1, 2, \ldots, n.$$

Therefore the Nash equilibrium points of $\mathcal{G}_2$ are the feasible solutions (i.e., real positive, if any) of (39) and the proof is completed. ■

Remark 2: It can be verified that for linear cost function for the players (i.e., $a_i = 0$), the Nash equilibrium point does not exist because in this case we have

$$N_i^\eta = \vartheta + \sum_{l\neq i} N_l^\eta, \qquad i = 1, 2, \ldots, n. \tag{40}$$

Summing up both sides of (40) results in

$$(2 - n)\sum_i N_i^\eta = n\vartheta$$

But $n \geq 2$ and $\vartheta > 0$.

B. Dynamic Estimation and Learning of $\mathcal{G}_2$

Putting (36) into (9) and rewriting (7) for $\mathcal{G}_2$, the players' estimation and decision making dynamics is obtained as follows:

$$\begin{aligned}\dot{\eta}_i =\ & -\frac{\gamma_i\eta_i}{\hat{Z}_i(\eta) + \eta_i}(\hat{\lambda}_i(\eta) - b_i)^2 \\ & \times \left((1 + 2a_i\hat{Z}_i(\eta))\eta_i - \hat{Z}_i(\eta)\right)\end{aligned} \tag{41a}$$

$$\dot{\eta}_{ij} = \delta_{ij}(\eta_j - \eta_{ij}), \qquad i, j = 1, 2, \ldots, n; \quad i \neq j \tag{41b}$$

where $\hat{Z}_i(\eta) = \vartheta + \sum_{l\neq i}\eta_{\mathrm{il}}$ and $\hat{\lambda}_i(\eta)$, the estimate of λ by the ith player, is obtained by replacing η_j, $j \neq i$, with its

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estimate η_{ij} in (3), i.e.,

$$\hat{\lambda}_i(\eta) = \frac{D_0 + \sum_j \beta_j \eta_{ij}}{\vartheta + \sum_j \eta_{ij}}. \tag{42}$$

Now, letting

$$\eta = (\eta_{1*}^T \dots \eta_{n*}^T)^T \in \mathbb{R}^{n^2} \tag{43}$$

it is straightforward proving what follows.

Proposition 6: There exists an equilibrium point $\eta^* = \underline{1}_n \otimes N^\eta$ for (41) corresponding to the Nash equilibrium of $\mathcal{G}_2$.

Notice that, with some abuse of notation, $\eta_i^* = \eta_{ii}^* = N_i^\eta$. Consider the following change of variables to shift the equilibrium point η^* to the origin:

$$y = \eta - \eta^*.$$

Subscripts on y will be the same as in (43).

Now, (41) can be rewritten as follows:

$$\begin{aligned}\dot{y}_i =& -\frac{\gamma_i(\eta_i^* + y_i)}{\hat{Z}_i(\eta^* + y) + (\eta_i^* + y_i)}[\hat{\lambda}_i(\eta^* + y) - b_i]^2 \\ &\times \left[\begin{array}{l}\left(1 + 2a_i\left(\vartheta + \sum_{j\neq i} y_{ij} + \sum_{j\neq i}\eta_j^*\right)\right)(y_i + \eta_i^*) \\ -\left(\vartheta + \sum_{j\neq i} y_{ij} + \sum_{j\neq i}\eta_j^*\right)\end{array}\right] \\ \dot{y}_{ij} =& \ \delta_{ij}(y_j - y_{ij}), \qquad i,j = 1,2,\dots,n; \quad i \neq j.\end{aligned} \tag{44}$$

After some simplifications related to η^* being an equilibrium point, we have

$$\begin{aligned}\dot{y}_i =& -\frac{\gamma_i(\eta_i^* + y_i)}{\hat{Z}_i(\eta^* + y) + (\eta_i^* + y_i)}[\hat{\lambda}_i(\eta^* + y) - b_i]^2 \\ &\times \left[\left(1 + 2a_i\left(\vartheta + \sum_{j\neq i} y_{ij} + \sum_{j\neq i}\eta_j^*\right)\right) y_i \right. \\ &\left. + \ (2a_i\eta_i^* - 1)\sum_{j\neq i} y_{ij}\right] \\ \dot{y}_{ij} =& \ \delta_{ij}(y_j - y_{ij}), \qquad i,j = 1,2,\dots,n; \quad i \neq j.\end{aligned} \tag{45}$$

We let

$$\kappa_{i1}(y) = \frac{\gamma_i(\eta_i^* + y_i)}{\hat{Z}_i(\eta^* + y) + \eta_i^* + y_i}[\hat{\lambda}_i(\eta^* + y) - b_i]^2 \tag{46a}$$

$$\kappa_{i2}(y) = 1 + 2a_i\left(\vartheta + \sum_{j\neq i} y_{ij} + \sum_{j\neq i}\eta_j^*\right) \tag{46b}$$

$$\kappa_{i3} = 1 - 2a_i\eta_i^* \tag{46c}$$

so that the estimation and decision making dynamics of suppliers can be written as

$$\dot{y}_i = -\kappa_{i1}(y)\kappa_{i2}(y)y_i + \kappa_{i1}(y)\kappa_{i3}\sum_{j\neq i} y_{ij} \tag{47a}$$

$$\dot{y}_{ij} = \delta_{ij}(y_j - y_{ij}), \qquad i,j = 1,2,\dots,n; \quad i \neq j \tag{47b}$$

where $\kappa_{i1}(y) \geq 0$, $\kappa_{i2}(y) > 1$. Also from (39), we have $0 < 2a_i\eta_i^* < 1$ and therefore $0 < \kappa_3 < 1$.

Remark 3: $\kappa_{i1}(y) = 0$ if and only if $y \in \mathcal{F}_i$ where

$$\mathcal{F}_i = \{y | y_i = -\eta_i^* \text{ or } \hat{\lambda}_i(y + \eta_i^*) = b_i\}. \tag{48}$$

Since $\eta_i^* \neq 0$ in view of (35) and, generically, $\hat{\lambda}_i(\eta^*) \neq b_i$, $\forall i$, we can assume $0 \notin \mathcal{F}_i$, i.e., $\kappa_{i1}(0) \neq 0$, $\forall i$.

C. Stability Analysis

Theorem 2: If the estimation rates δ_{ij} and the learning rates γ_i are appropriately chosen, the dynamic system (47) is locally asymptotically stable at the origin. An estimate of the region of attraction is given by the ball $E_r = \{y | \|y\| < r\}$, with r such that $E_r \cap \{\mathcal{F}_1 \cup \mathcal{F}_2 \cup \dots \cup \mathcal{F}_n\} = \emptyset$.

Proof: Consider the following Lyapunov function for system (47):

$$V(y) = \frac{1}{2}\sum_i\left(y_i^2 + \sum_{j\neq i} y_{ij}^2\right) = \frac{1}{2}\sum_i\left(y_i^2 + \sum_{j\neq i} y_{ji}^2\right). \tag{49}$$

Then we have

$$\begin{aligned}\dot{V} =& \sum_i\left(\begin{array}{l}-\kappa_{i1}(y)\kappa_{i2}(y){y_i}^2 + \kappa_{i1}(y)\kappa_{i3}\ y_i\sum_{j\neq i} y_{ij} \\ + \sum_{j\neq i} y_{ji}\delta_{ij}(y_i - y_{ij})\end{array}\right) \\ =& \sum_i\left(\begin{array}{l}-\kappa_{i1}(y)\kappa_{i2}(y){y_i}^2 + \kappa_{i1}(y)\kappa_{i3}\ y_i\sum_{j\neq i} y_{ij} \\ +y_i\sum_{j\neq i}\delta_{ji}y_{ji} - \sum_{j\neq i}\delta_{ji}y_{ji}^2\end{array}\right) \\ =& \sum_i\left(\begin{array}{l}-\kappa_{i1}(y)\kappa_{i2}(y){y_i}^2 - \sum_{j\neq i}\delta_{ji}y_{ji}^2 \\ +y_i\left(\kappa_{i1}(y)\kappa_{i3}\sum_{j\neq i} y_{ij} + \sum_{j\neq i}\delta_{ji}y_{ji}\right)\end{array}\right).\end{aligned} \tag{50}$$

Now, using the fact that $-\kappa_{i1}(y)\kappa_{i2}(y){y_i}^2 = \sum_{j\neq i} -\frac{1}{2(n-1)}\kappa_{i1}(y)\kappa_{i2}(y){y_i}^2 + \sum_{j\neq i} -\frac{1}{2(n-1)}\kappa_{i1}(y)\kappa_{i2}(y){y_i}^2$, we write (50) as

$$\dot{V} = \sum_i\sum_{j\neq i}\left(\theta_{ij}^T\Upsilon_{ij}\theta_{ij} + \sigma_{ij}^T\Phi_{ij}\sigma_{ij}\right) \tag{51}$$

where

$$\begin{aligned}\theta_{ij} =& \ (y_i \quad y_{ij})^T, \\ \sigma_{ij} =& \ (y_i \quad y_{ji})^T \\ \Upsilon_{ij} =& \begin{pmatrix}-\frac{1}{2(n-1)}\kappa_{i1}(y)\kappa_{i2}(y) & \frac{1}{2}\kappa_{i1}(y)\kappa_{i3} \\ \frac{1}{2}\kappa_{i1}(y)\kappa_{i3} & -\frac{1}{2}\delta_{ij}\end{pmatrix} \\ \Phi_{ij} =& \begin{pmatrix}-\frac{1}{2(n-1)}\kappa_{i1}(y)\kappa_{i2}(y) & \frac{1}{2}\delta_{ji} \\ \frac{1}{2}\delta_{ji} & -\frac{1}{2}\delta_{ji}\end{pmatrix}.\end{aligned}$$

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Therefore, system (47) is asymptotically stable at the origin if both Υ and Φ are negative definite in a neighborhood of the origin, i.e.,

$$\frac{1}{2(n-1)}\kappa_{i1}(y)\kappa_{i2}(y) > 0 \tag{52a}$$

$$\frac{\delta_{ij}}{4(n-1)}\kappa_{i1}(y)\kappa_{i2}(y) - \frac{1}{4}\kappa_{i1}^2(y)\kappa_{i3}^2 > 0 \tag{52b}$$

$$\frac{\delta_{ji}}{4(n-1)}\kappa_{i1}(y)\kappa_{i2}(y) - \frac{1}{4}\delta_{ji}^2 > 0. \tag{52c}$$

Since $\kappa_{i1}(0) \neq 0$, $\kappa_{i1}(y) > 0$ in E_r; further, since $\kappa_{i2}(y) > 1$ the condition (52a) is satisfied. Also by replacing $\kappa_{i2}(y)$ from (46b) and putting into (52b) and (52c), we get the conditions

$$a_i > \frac{1}{2\left(\vartheta + \sum_{j\neq i} y_{ij} + \sum_{j\neq i}\eta_j^*\right)}\left(\frac{(n-1)\kappa_{i1}(y)\kappa_{i3}^2}{\delta_{ij}} - 1\right)$$

$$a_i > \frac{1}{2\left(\vartheta + \sum_{j\neq i} y_{ij} + \sum_{j\neq i}\eta_j^*\right)}\left(\frac{(n-1)\delta_{ji}^2}{\kappa_{i1}(y)} - 1\right) \tag{53}$$

which are certainly satisfied if the rhs's are negative, i.e.,

$$(n-1)\delta_{ji}^2 < \kappa_{i1}(y) < \frac{\delta_{ij}}{(n-1)\kappa_{i3}^2}. \tag{54}$$

In view of (46a), we can write $\kappa_{i1}(y) = \gamma_i\tilde{\kappa}_{i1}(y)$, with $\kappa_{i1}(\cdot)$ continuous, so that

$$\gamma_i\tilde{\kappa}_{i1}^{\min} \leq \kappa_{i1}(y) \leq \gamma_i\tilde{\kappa}_{i1}^{\max}$$

with $\tilde{\kappa}_{i1}^{\max} = \max_{y\in E_r}\tilde{\kappa}_{i1}(y)$ and $\tilde{\kappa}_{i1}^{\min} = \min_{y\in E_r}\tilde{\kappa}_{i1}(y)$. Now (54) becomes

$$\frac{(n-1)\delta_{ji}^2}{\tilde{\kappa}_{i1}^{\min}} < \gamma_i < \frac{\delta_{ij}}{(n-1)\kappa_{i3}^2\tilde{\kappa}_{i1}^{\max}}. \tag{55}$$

Choosing the parameters δ_{ij} such that $\delta_{\min} < \delta_{ij} < \nu\delta_{\min}$, $\forall i, j$ with $\nu > 1$, the satisfaction of (55)—and, hence, the asymptotic stability of the statement—is guaranteed by choosing

$$\delta_{\min} < \frac{\tilde{\kappa}_{i1}^{\min}}{(n-1)^2\nu^2\kappa_{i3}^2\tilde{\kappa}_{i1}^{\max}}. \tag{56}$$

■

Remark 4: It is not difficult to realize that the demand uncertainty does not affect the Nash equilibrium of $\mathcal{G}_2$ or the corresponding equilibrium point of (47). Nevertheless, it affects the region of attraction by changing $\mathcal{F}_i$ and $\kappa_{i1}(y)$.

V. Case Study

In this study, a market consisting of three suppliers is considered. First, we consider suppliers to compete in $\mathcal{G}_1$ and thus, the intercept of the offering supply curve is the strategy of the players. In the second case, the slope of the supply function is the strategy of the players, while the intercept is considered as the corresponding marginal cost coefficient.

TABLE I
Parameters of the Players in $\mathcal{G}_1$

Parameter	$\eta_i(1)$	$\eta_i(2)$	a_i	b_i	c_i	γ_i	δ_{ij}
S1	1	10/3	0.4	8	0	0.2	3
S2	10/11	100/33	0.5	10	0	0.2	3
S3	5/6	25/9	0.6	12	0	0.2	3

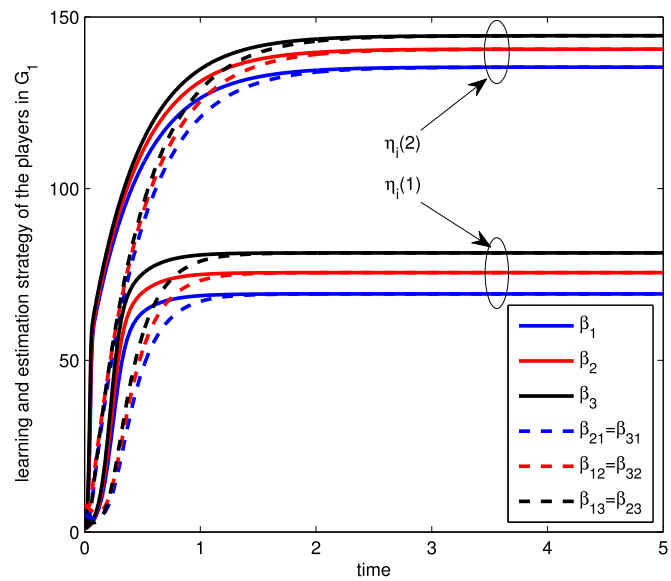

Fig. 2. Players' learning and estimation strategies and convergence to Nash equilibrium in different slopes.

A. Results of $\mathcal{G}_1$

The parameters of the estimation and decision making of the players are given in Table I. The base demand D_0 and demand elasticity ϑ are set equal to 500 and 0.1, respectively. The game parameters and different slope parameter sets (i.e., $\{\eta_i(1)\}, \{\eta_i(2)\}$) are denoted in Table I.

Fig. 2 shows the effect of variation in slope of supply function on the stability of the system. It can be seen that in the case $\eta_i(1)$, the dominant eigenvalue of A have larger negative real value (i.e., $-2.28 > -1.14$) and this leads to faster convergence to the equilibrium point. In addition, we can see that by increasing the slopes (approximately three times in this example), since the price is decreased [based on (3)] and as a result demand is increased [based on (2)], therefore the Nash equilibrium strategies (intercepts) is increased to satisfy the demand and supply balance.

To study the robustness of the proposed learning and estimation method in presence of disturbance, we consider a disturbance $d(t) = 10\sin(t)$ in (32). Although the new system does not have any equilibrium point, but we are interested to figure out whether the system is ultimately bounded or not and to what region system's trajectories ultimately belong. Fig. 3 shows that the system is ultimately bounded in presence of disturbance and converges close to the equilibrium point of the old system without disturbance.

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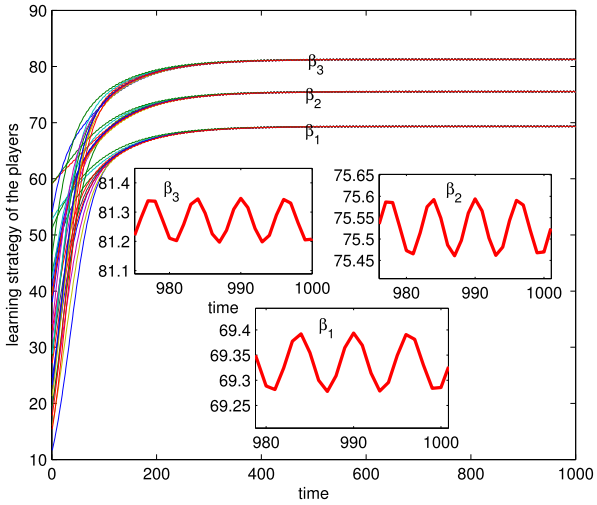

Fig. 3. Robustness against demand uncertainty in $\mathcal{G}_1$.

TABLE II
PARAMETERS OF THE PLAYERS IN $\mathcal{G}_1$

Parameter	$a_i(1)$	$a_i(2)$	$a_i(3)$	$b_i = \beta_i$	c_i	γ_i	δ_{ij}
S1	0.004	0.01	0.04	8	0	0.2	3
S2	0.005	0.0125	0.05	10	0.	0.2	3
S3	0.006	0.015	0.06	12	0	0.2	39

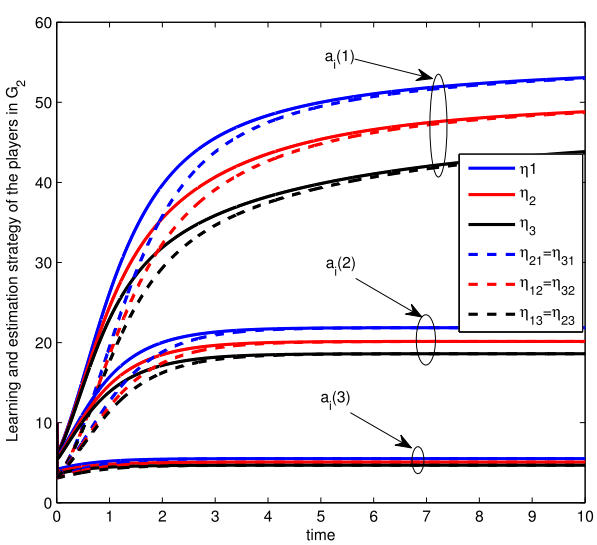

Fig. 4. Players' learning and estimation strategies under different cost coefficients.

B. Results of $\mathcal{G}_2$

The parameters of the players in this simulation is given in Table II. The base load is fixed to 500 and the demand elasticity is set to 0.1.

One can see from Fig. 4 that the learning and estimation strategies converge to the Nash equilibrium point of $\mathcal{G}_2$. By increasing a_i the Nash equilibrium point approaches the zero equilibrium point which is due to the fact that $a_i\eta_i^* < 1$.

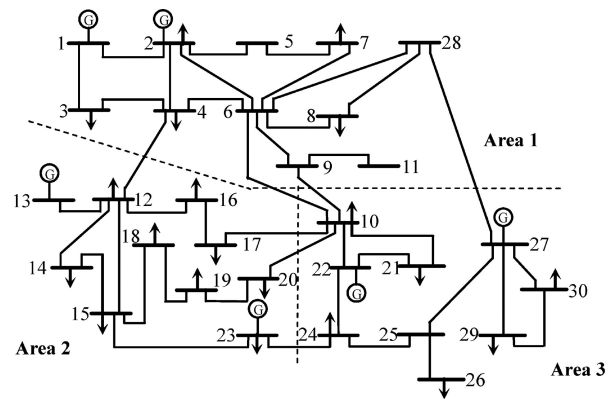

Fig. 5. IEEE 30-bus power system (with three interconnected demand and generation zones).

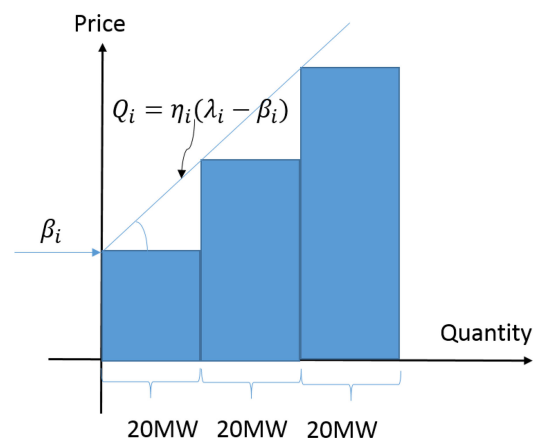

Fig. 6. Bid function.

C. Case Study on IEEE 30-bus Power System

To show the ability of the proposed method in the large scale systems and also realistic situations, a modified IEEE 30-bus system is studied which is shown in Fig. 5. The details of the system's model is available in "t_auction_case.m" at MATLAB Power System Simulation Package (MATPOWER 6.0b1), http://www.pserc.cornell.edu/matpower/. This power system has six generators and is divided into three demand areas (zones) with two generators in each area. It is assumed that each generator submits a supply function to the energy market. After collecting the bid functions, a dc OPF solver in MATPOWER toolbox is used to clear the market. In the market clearing process, the output power dispatches, energy prices, and transmission flows for all agents are determined based on the system constraints. The nodal prices (LMPs), which are equal to the prices of the marginal blocks are determined by the Run Market and dc OPF modules of the MATPOWER software. In order to examine the performance of the proposed learning and estimation procedure (9) for $\mathcal{G}_1$ and $\mathcal{G}_2$, we studied both of the games separately. Since, the system's constraints are ignored in the analysis, we first design the strategies of the players without constraints and then we apply it to the real system to see the performance. The cost function parameters and total demand are adopted from the IEEE standard 30-bus system (case30.m

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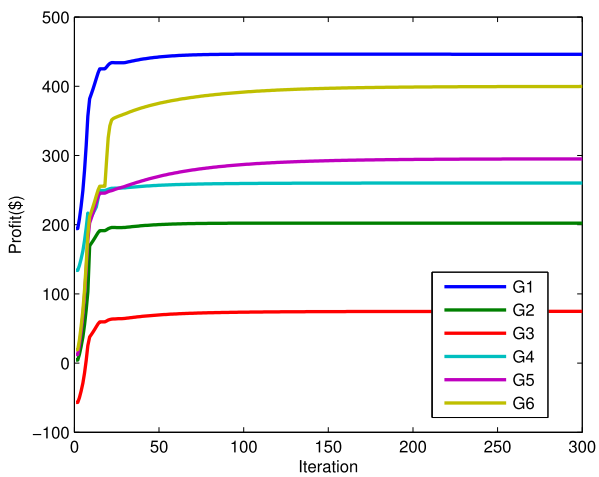

Fig. 7. Convergence of the sellers' profit maximizing learning process in $\mathcal{G}_1$.

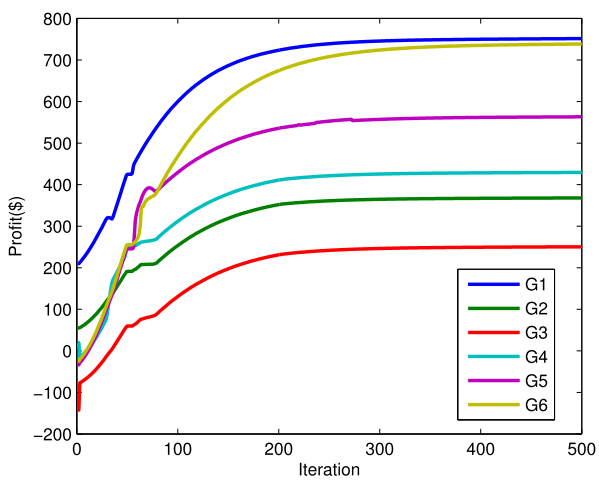

Fig. 8. Convergence of the sellers' profit maximizing learning process in $\mathcal{G}_2$.

available at http://www.pserc.cornell.edu/matpower/). The learning parameters γ_i and δ_i is set to 2 and 3 for all the players in both $\mathcal{G}_1$ and $\mathcal{G}_2$, respectively. ν is set equal to zero. In both of the cases $\mathcal{G}_1$ and $\mathcal{G}_2$, we start the offering process with a piecewise bid function of three blocks as shown in Fig. 6. After finding the new strategy in each iteration, we adjust the bid function by updating the β_i and η_i in $\mathcal{G}_1$ and $\mathcal{G}_2$, respectively. The profits of six generators under $\mathcal{G}_1$ are shown in Fig. 7. We can see that using the proposed dynamic estimation and learning (12), players's strategies and corresponding profits converges after about 50 iterations and they are able to learn to increase their profit in a repeated game. The same results has been found after studying $\mathcal{G}_2$ after 200 iterations by using (41) as shown in Fig. 8. It is an expected result that the learning process of the slope of supply function takes more time compared to the intercept because of the nonlinearity in the best response functions in $\mathcal{G}_2$.

VI. Conclusion

In this paper, the competition among the suppliers in a uniform price market under SFE mechanism was modeled as an incomplete information game. Players tried to learn their optimal strategy gradually, in a continuous time repeated game setting, using a bounded rational decision making and taking into account the decisions of rivals by an estimator. The learning and estimation dynamics of the players were studied in separate game problems (i.e., $\mathcal{G}_1$ and $\mathcal{G}_2$) where either the slope or the intercept of the linear supply function were considered as the strategy of the players, respectively. In each case, the Nash equilibrium point of the game was derived and it was shown that there exists a corresponding equilibrium point in estimation and learning dynamics. Then, the stability of such an equilibrium point was studied and an estimate of domain of attraction was computed. The effects of demand uncertainty on system's dynamic was also analyzed employing the concept of practical stability.

In particular, we found out that for the game $\mathcal{G}_1$, the asymptotic stability of the equilibrium point first of all depends on the parameters of demand, cost, and supply functions of the players; if some sufficient conditions are satisfied on those parameters, then the players have a chance to converge to the corresponding Nash equilibrium point by a suitable selection of the estimation and learning rates; an estimate of the attraction domain was also provided. As regards $\mathcal{G}_2$, if the estimation and learning rates are chosen appropriately, then the system is always locally asymptotically stable. In addition, we showed that the demand uncertainty could change both the attractor and the domain of attraction in the case of $\mathcal{G}_1$; instead, for $\mathcal{G}_2$, demand uncertainty does not change the equilibrium point of system's dynamics but does affect the attraction domain.

The future research on this paper could be devoted on the extension of quadratic cost function to general convex cost functions. Also the analysis would become nearer to applications if one could perform the convergence and stability analysis in presence of the system's constraints such as price and generation/demand limits. In addition, analytical analysis (convergence and stability) on simultaneous learning of slope and intercept of supply function is an open problem. The extension of framework to the double sided auctions where the buyers also can submit a bid function to the ISO would be another interesting study. The uniform price auction is studied in this paper. Other market clearing mechanisms such as first price, second price, and pay-as bid auctions could be studied as well.

Appendix

A. First Derivative of Payoff Function for $\mathcal{G}_1$

The first derivative of payoff function with respect to β_i is

$$\frac{\partial \pi_i}{\partial \beta_i} = \frac{\partial \lambda}{\partial \beta_i} Q_i + \frac{\partial Q_i}{\partial \beta_i} \lambda - 2a_i Q_i \frac{\partial Q_i}{\partial \beta_i} - b_i \frac{\partial Q_i}{\partial \beta_i}. \tag{57}$$

Considering $q_i = \frac{\eta_i}{\vartheta + \sum_j \eta_j}$ we have

$$\frac{\partial \lambda}{\partial \beta_i} = q_i \tag{58}$$

and

$$\frac{\partial Q_i}{\partial \beta_i} = \frac{\partial}{\partial \beta_i} \left(\eta_i (\lambda - \beta_i)\right) = \eta_i \left(q_i - 1\right). \tag{59}$$

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Replacing (58) and (59) into (57) and using the expression of λ from (3), we finally get

$$\begin{aligned}\frac{\partial \pi_i}{\partial \beta_i} = & \ \beta_i \eta_i \left[(q_i - 1)\left[q_i - 2a_i\eta_i\left(q_i - 1\right)\right] + q_i\left(q_i - 1\right)\right] \\ & + q_i\left[q_i - 2a_i\eta_i\left(q_i - 1\right) + \left(q_i - 1\right)\right]\sum_{j\neq i}\beta_j\eta_j \\ & + \left[D_0 q_i\left[q_i + \left(1 - 2a_i\eta_i\right)\left(q_i - 1\right)\right] - b_i\eta_i\left(q_i - 1\right)\right]. \end{aligned} \tag{60}$$

Now by taking the second derivative with respect to β_i, we have

$$\frac{\partial^2 \pi_i}{\partial {\beta_i}^2} = \eta_i\left[(q_i - 1)\left[q_i - 2a_i\eta_i\left(q_i - 1\right)\right] + q_i\left(q_i - 1\right)\right] \tag{61}$$

which can be easily written as (10).

B. Nash Equilibrium Point of $\mathcal{G}_1$

Using (60), the Nash equilibrium point of $\mathcal{G}_1$ is given by (11), where

$$\begin{aligned} & w : w_i = \tfrac{b_i}{q_i}(q_i - 1) - \tfrac{D_0}{K}(1 + \tfrac{1}{q_i}(q_i - 1)(1 - 2a_i\eta_i)) \\ & i = 1, 2, \ldots, n \end{aligned}$$

$$Y : \begin{cases} Y_{ij} = & (2q_j - \frac{q_j}{q_i} - 2\frac{q_j}{q_i}a_i\eta_i(q_i - 1)), \ \ i \neq j \\ Y_{ii} = & 2(q_i - 1)(1 - \frac{1}{q_i}a_i\eta_i(q_i - 1)), \ i, j = 1, \ldots, n. \end{cases}$$

C. Coefficients of Dynamic Learning in $\mathcal{G}_1$

According to (60), the coefficients of (12a) are obtained as follows:

$$h_{ij} = \eta_j\left[q_i^2 + q_i\left(1 - 2a_i\eta_i\right)\left(q_i - 1\right)\right] \tag{62a}$$

$$f_i = 2(q_i - 1)\eta_i\left(q_i - a_i\eta_i(q_i - 1)\right) \tag{62b}$$

$$\begin{aligned} m_i = & \ D_0\left[q_i^2 + q_i\left(1 - 2a_i\eta_i\right)\left(q_i - 1\right)\right] \\ & - b_i\eta_i\left(q_i - 1\right). \end{aligned} \tag{62c}$$

D. Proof of Proposition 3

Proof: Consider the following Lyapunov function for the system (17):

$$V(x) = x^T Hx > 0\,. \tag{63}$$

Consequently, we have

$$\begin{aligned} \dot{V} &= \dot{x}^T \mathrm{Hx} + x^T H\dot{x} \\ &= \left[\mathrm{Ax} + \Sigma(x)\mathrm{Bx}\right]^T \mathrm{Hx} + x^T H\left[\mathrm{Ax} + \Sigma(x)\mathrm{Bx}\right] \\ &= x^T\left(A^T H + HA^T\right)x + x^T\left[B^T\Sigma(x)H + H\Sigma(x)B\right]x \\ &= -x^T\mathrm{Gx} + 2x^T B^T\Sigma(x)\mathrm{Hx}. \end{aligned} \tag{64}$$

Then we have

$$\begin{aligned} \dot{V} &\leq -x^T\mathrm{Gx} + 2\|B\|\|\Sigma(x)\|\|H\|\,\|x\|^2 \\ &\leq \Big[-\lambda_{\min}(G) + 2\|B\|\|\Sigma(x)\|\|H\|\Big]\|x\|^2. \end{aligned} \tag{65}$$

Since

$$\|B\| = \max_i \|B_{ii}\| = \max_i \gamma_i\sqrt{f_i^2 + \sum_{j\neq i} h_{ij}^2}$$

and

$$\|\Sigma(x)\| = \sqrt{\lambda_{\max}(\Sigma(x)^2)} = \max_i |x_i| \leq \|x\| \tag{66}$$

by using (31) and (65), we obtain that for sufficiently small r

$$\dot{V} < 0$$

and the proof is completed. ■

E. Proof of Proposition 4

Proof: We take a Lyapunov function as

$$V(x) = x^T\mathrm{Hx}.$$

We then have

$$\begin{aligned} \dot{V} = & \ \dot{x}^T\mathrm{Hx} + x^T H\dot{x} \\ = & \ [\mathrm{Ax} + \Sigma(x)\mathrm{Bx} + S(\beta^* + x)d]^T\mathrm{Hx} \\ & + x^T H[\mathrm{Ax} + \Sigma(x)\mathrm{Bx} + S(\beta^* + x)d] \\ = & \ x^T\Big[(A + \Sigma(x)B)^T H \\ & + H(A + \Sigma(x)B)\Big]x + 2dx^T H\mathrm{Sx} + 2dx^T\mathrm{HS}\beta^*. \end{aligned}$$

Now according to Proposition 3 and (65), in the domain specified by U_ϵ we have

$$x^T\left[(A + \Sigma(x)B)^T H + H(A + \Sigma(x)B)\right]x \leq -\xi_\epsilon\|x\|^2$$

where $\xi_\epsilon = -\lambda_{\min}(G) + 2\epsilon\|B\|\|H\|$, and therefore

$$\dot{V} \leq -\xi_\epsilon\|x\|^2 + 2d\left(x^T H\mathrm{Sx} + x^T\mathrm{HS}\beta^*\right).$$

According to triangular norm inequalities, we have

$$\dot{V} \leq -\xi_\epsilon\|x\|^2 + 2\bar{d}\,\|H\|\|S\|(\|x\|^2 + \|\beta^*\|\|x\|).$$

Since $H > 0$ and S is diagonal

$$\|H\| = \lambda_{\max}(H) \quad \|S\| = \max_i(\gamma_i\,|M_i|).$$

Now if we take $\zeta = \|H\|\,\|S\|$, we have

$$\begin{aligned} \dot{V} &\leq -\xi_\epsilon\|x\|^2 + 2\bar{d}\zeta(\|x\|^2 + \|x\|\,\|\beta^*\|) \\ &= (2\bar{d}\zeta - \xi_\epsilon)\|x\|^2 + 2\bar{d}\zeta\|x\|\|\beta^*\|. \end{aligned}$$

Therefore a necessary condition for negativeness of $\dot{V}$ is

$$\bar{d} < \frac{\xi_\epsilon}{2\zeta}$$

consequently, $\dot{V}$ is negative if

$$\|x\| > \frac{2\bar{d}\zeta\,\|\beta^*\|}{(\xi_\epsilon - 2\bar{d}\zeta)} = \upsilon(\epsilon, \bar{d}). \tag{67}$$

We need also to have $\upsilon < \epsilon$. Then according to (67), we obtain

$$\bar{d} < \frac{\xi_\epsilon\epsilon}{2\zeta(\|\beta^*\| + \epsilon)} = d_0(\epsilon). \tag{68}$$

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Once d_0 is found, $\dot{V}$ is negative in I where

$$I = \{x \mid \upsilon < \|x\| < \epsilon\}. \tag{69}$$

Therefore, the trajectories which initialize in I will go toward the origin and finally reach U_v. Also, all the trajectories inside $U_v = U_\epsilon - I$ will stay there. Consequently, all trajectories starting at U_ϵ will converge to U_v and the proof is completed. ■

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